

Recent studies in AI

Nguyễn Ngọc Thảo – Nguyễn Hải Minh
{nnthao, nhminh}@fit.hcmus.edu.vn

Generative AI

- Generative AI refers to **AI models** that are able to **generate new, original content** using **generative models**.

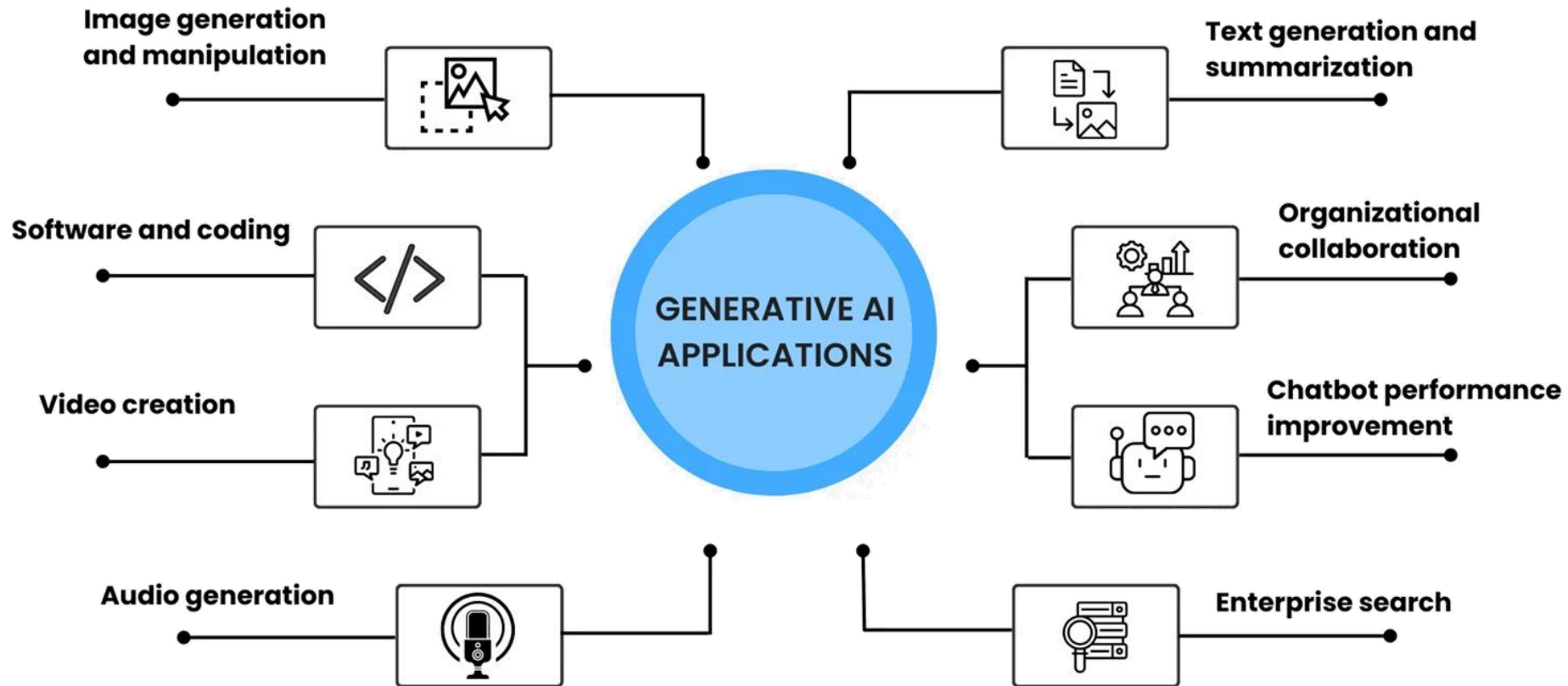


Prompt

A cute corgi lives in a house made out of sushi.



Generative AI



Text generation and summarization

- **Text generation** finds applications in various fields, including content creation, chatbots, news summarization, and more.
 - It is a dynamic area of research and development in the broader field of Natural language processing and Artificial intelligence.



ChatGPT



Bing

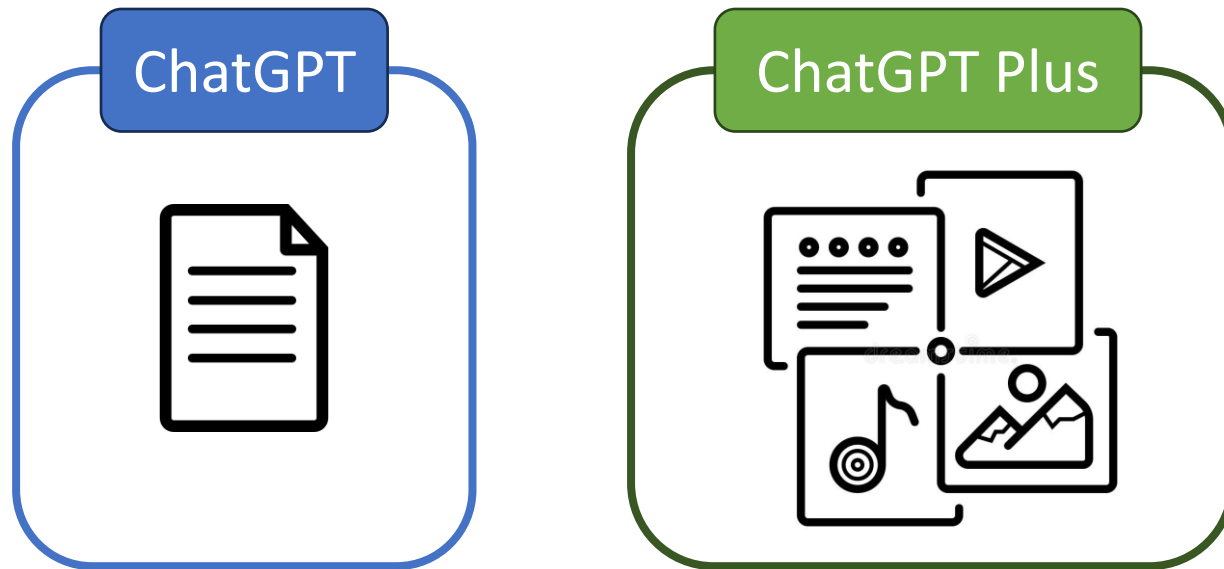


Bard

- Most of the tools rely on **large language models** (LLMs).

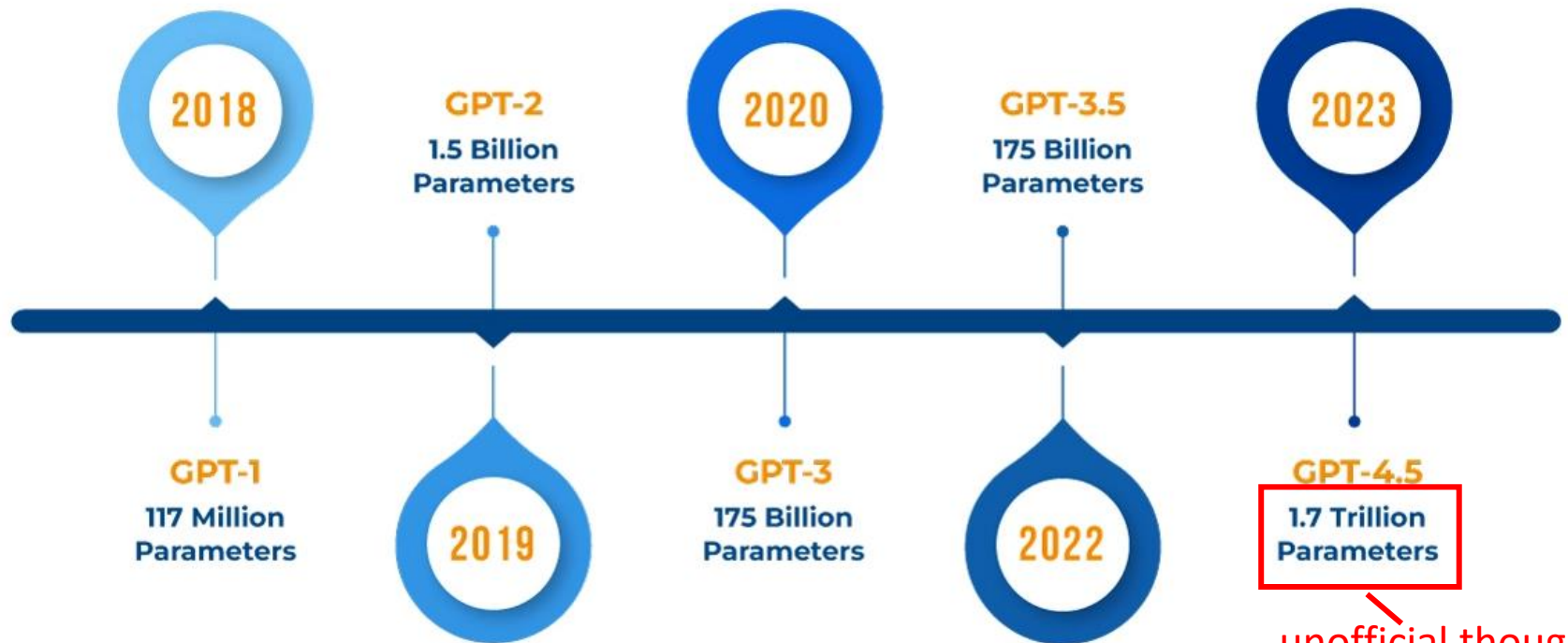
ChatGPT: An introduction

- ChatGPT is built upon either GPT-3.5 or GPT-4, that were fine-tuned to target conversational usage.
 - It leverages both supervised learning and reinforcement learning in a process called reinforcement learning from human feedback (RLHF).



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unofficial though 6

Transformer:

The key technique

- Transformer is a versatile neural network architecture for both unsupervised and supervised learning tasks.

Vaswani, Ashish, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. "Attention is all you need." *Advances in neural information processing systems* 30 (2017).

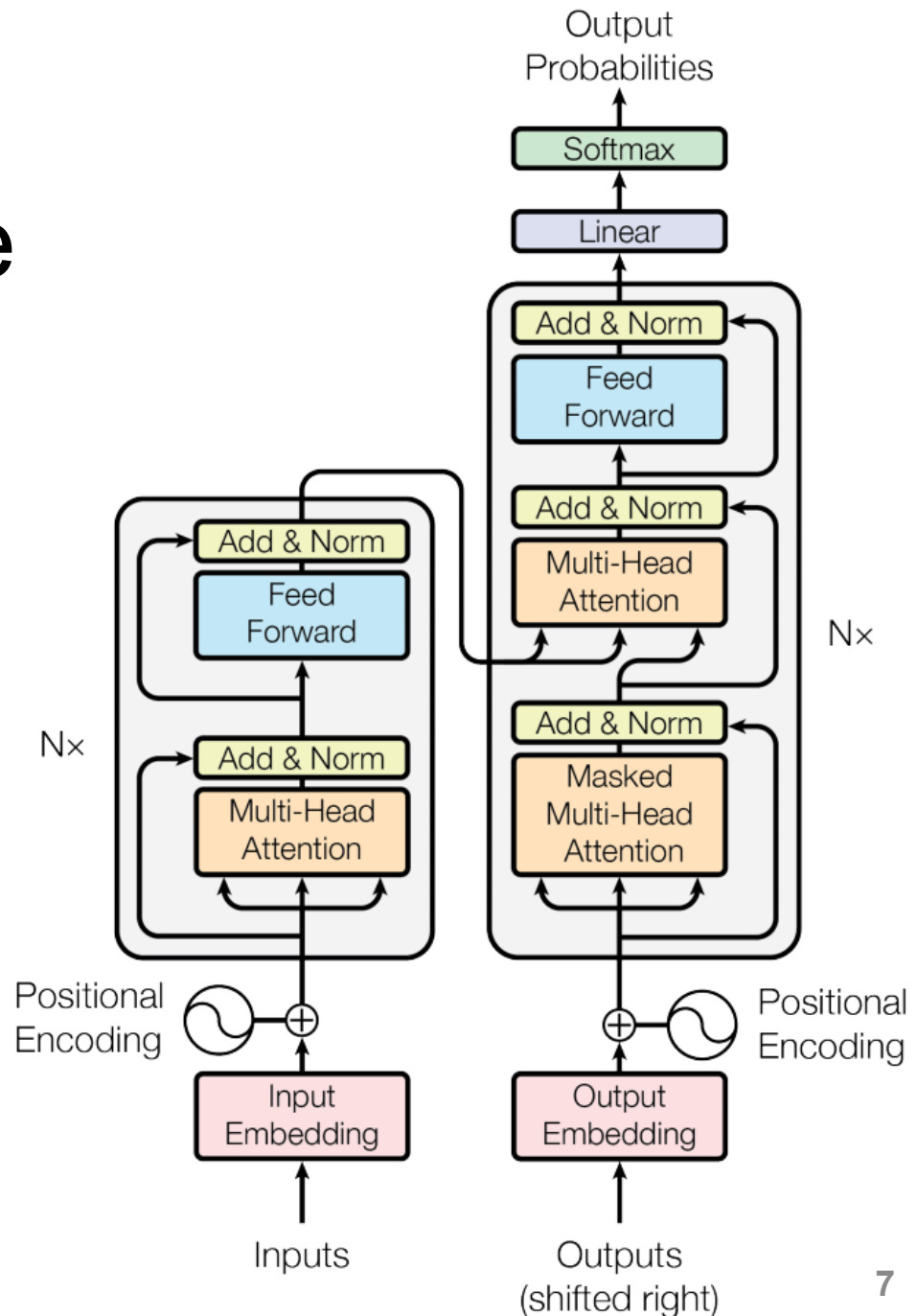


Image generation and manipulation

- **Image generation** include art creation, data augmentation for ML models, creating realistic scenes for movies, and more.
 - It is an evolving field with continuous advancements in both traditional and machine learning-based approaches.



StyleGAN



Imagen

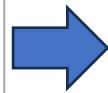


DALL-E

Text-to-image translation

- The task aims to generate an image that corresponds to a given textual description.

A modern architectural building with large glass windows, situated on a cliff overlooking a serene ocean at sunset



Meaningful
representation



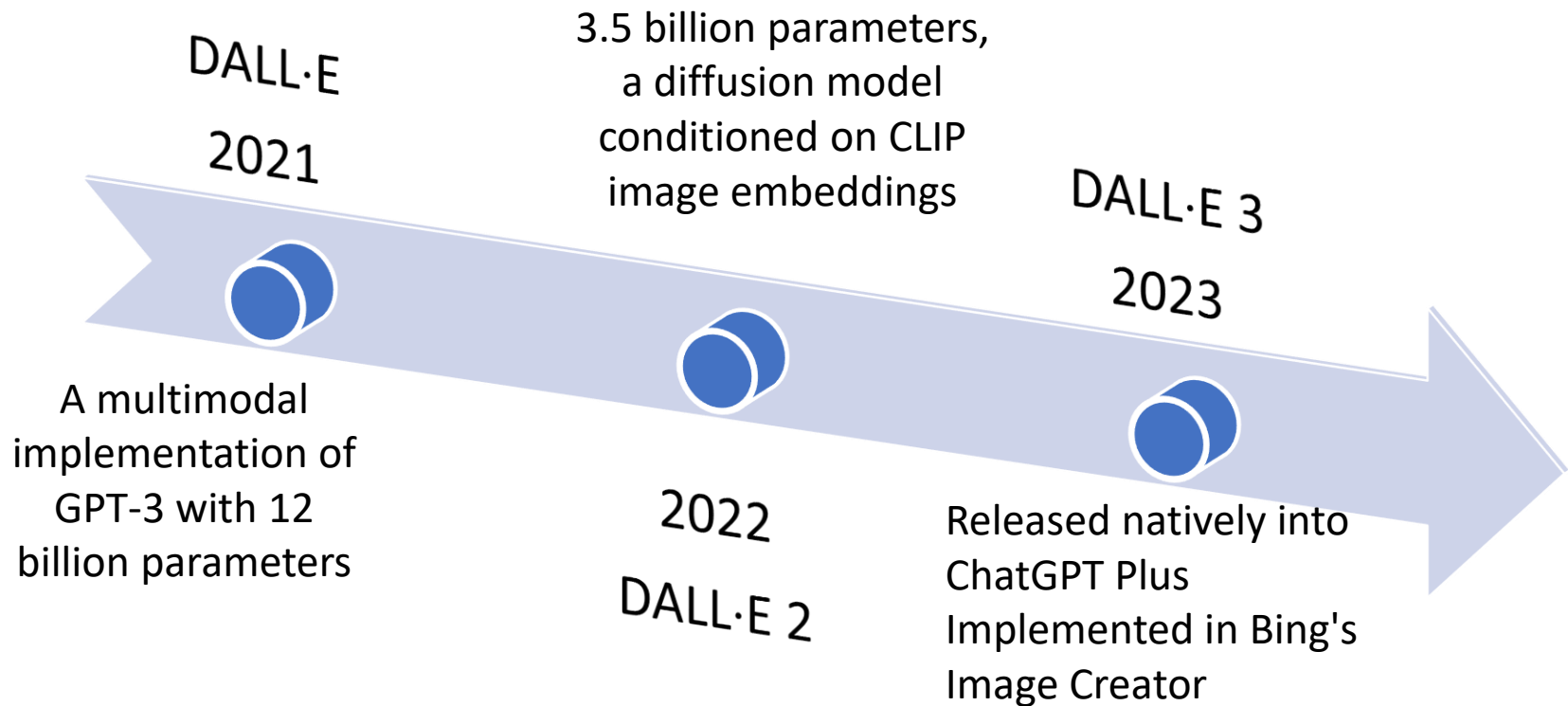
Created by DALL·E 3

- It is a reverse task to image captioning.

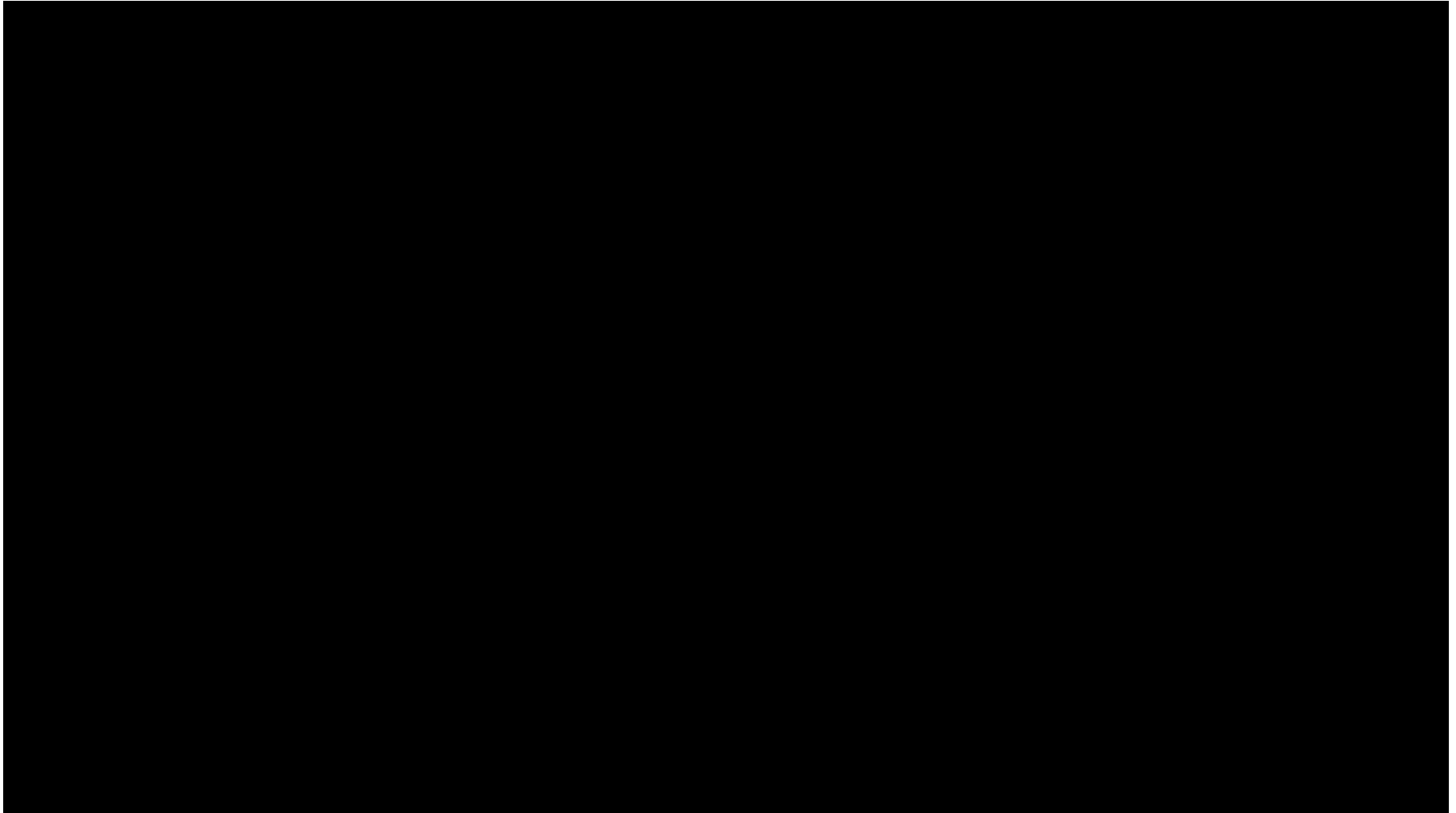
DALL·E: An introduction



- DALL·E is a **text-to-image model using deep learning** to generate digital images from natural language prompts.

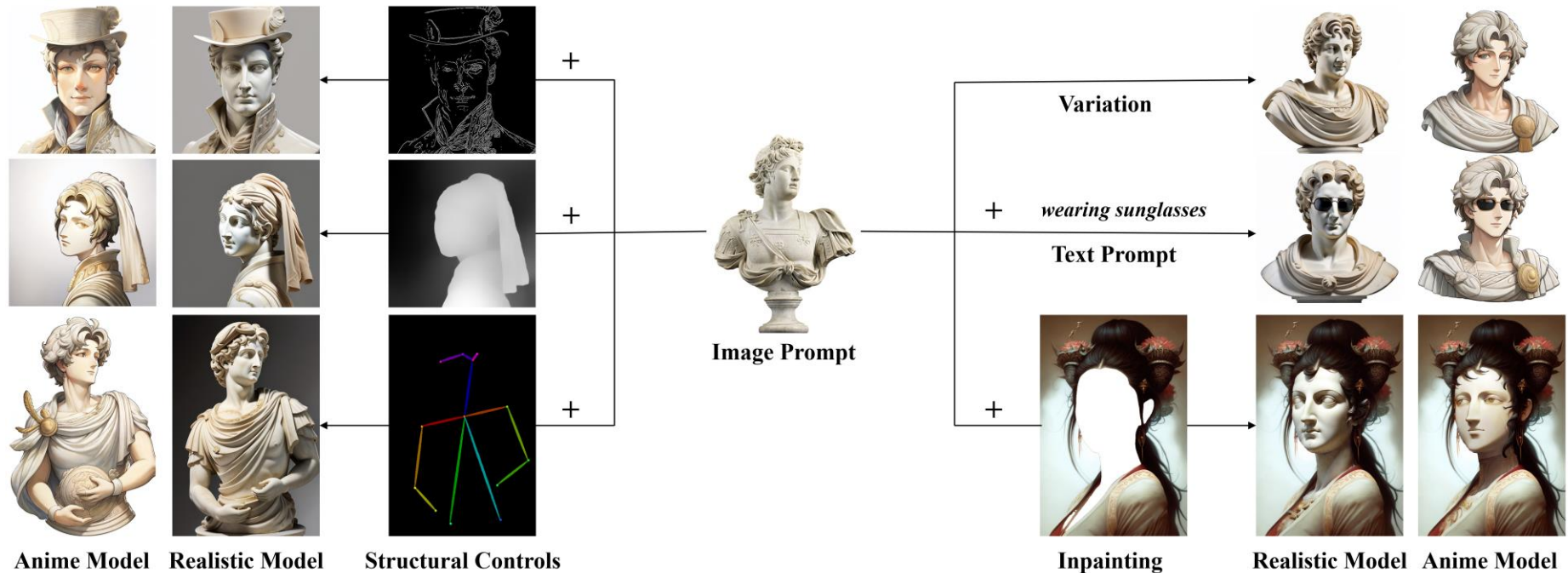


DALL·E 2: Demonstration



Source: [YouTube video](#)

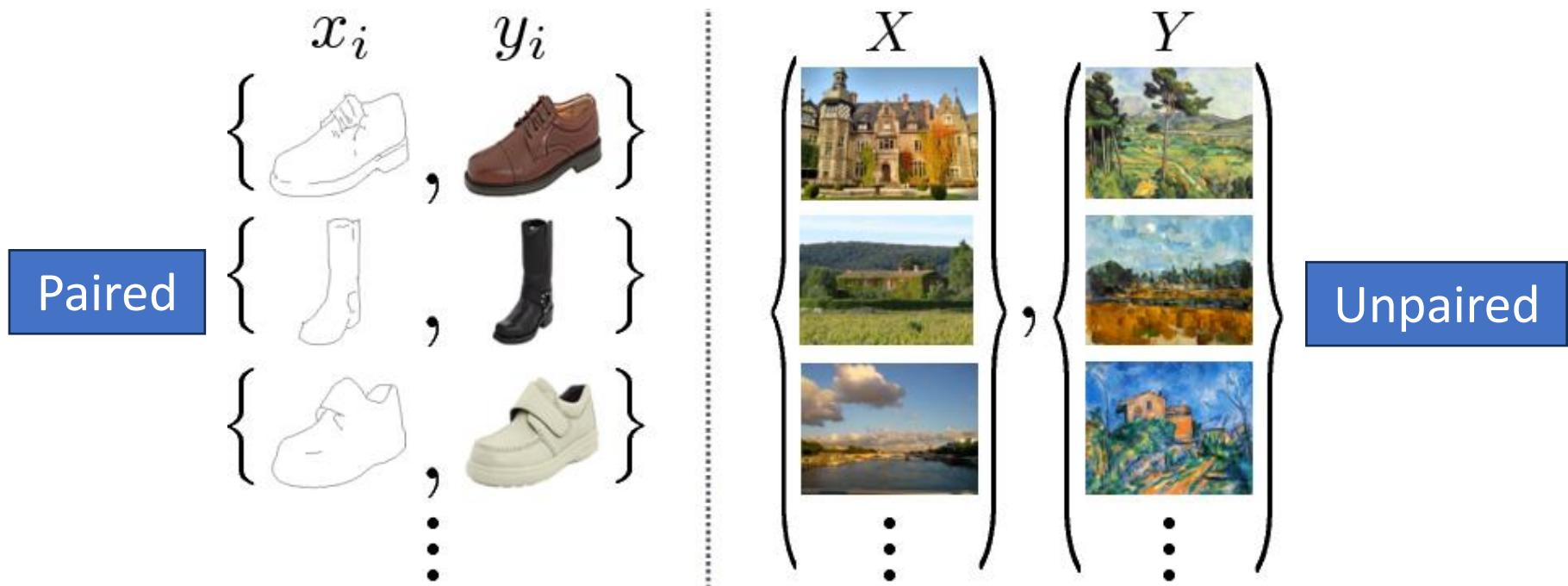
Image generation with prompts



Various image synthesis with [IP-Adapter](#) applied on the pretrained text-to-image diffusion models with different styles

Image-to-image translation

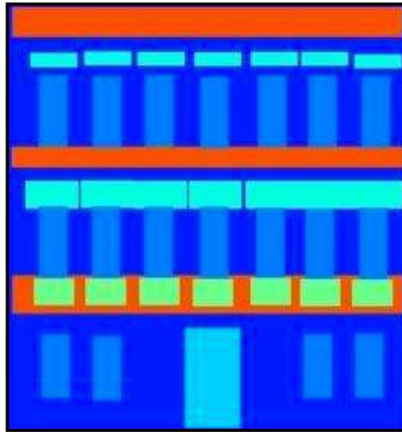
- The task is the process of transforming an image from one domain to another.



Paired training data (left) consists of training examples have one to one correspondence. Unpaired training set has no such correspondence.

Image-to-image translation tasks

Labels to Facade



input



output

BW to Color



input



output

Day to Night



input



output

Edges to Photo

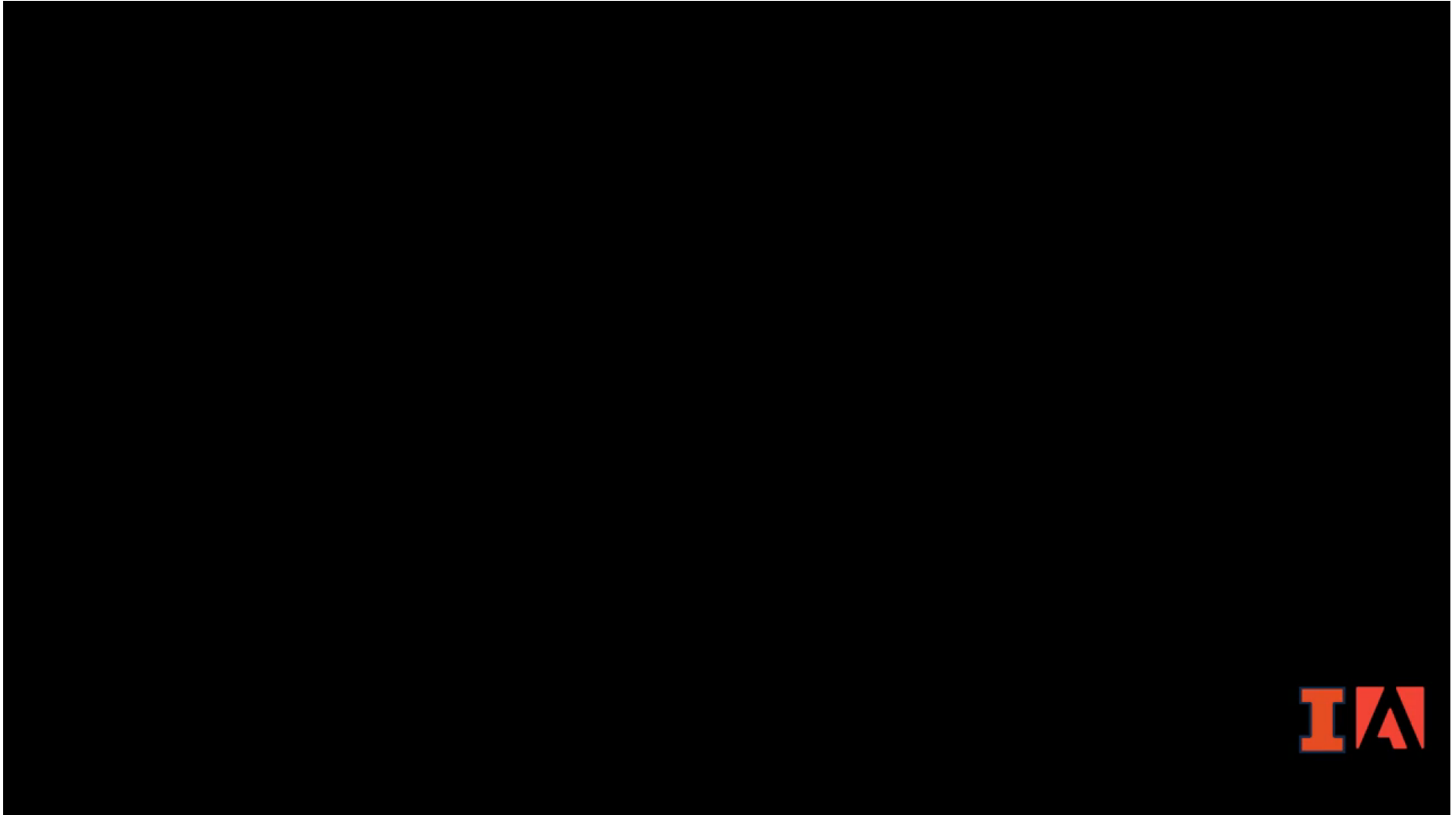


input



output

Image inpainting using GAN



Zhou, Yuqian, et al. "TransFill: Reference-guided Image Inpainting by Merging Multiple Color and Spatial Transformations." Proceedings of the IEEE/CVF CVPR. 2021.

Image synthesis using GAN

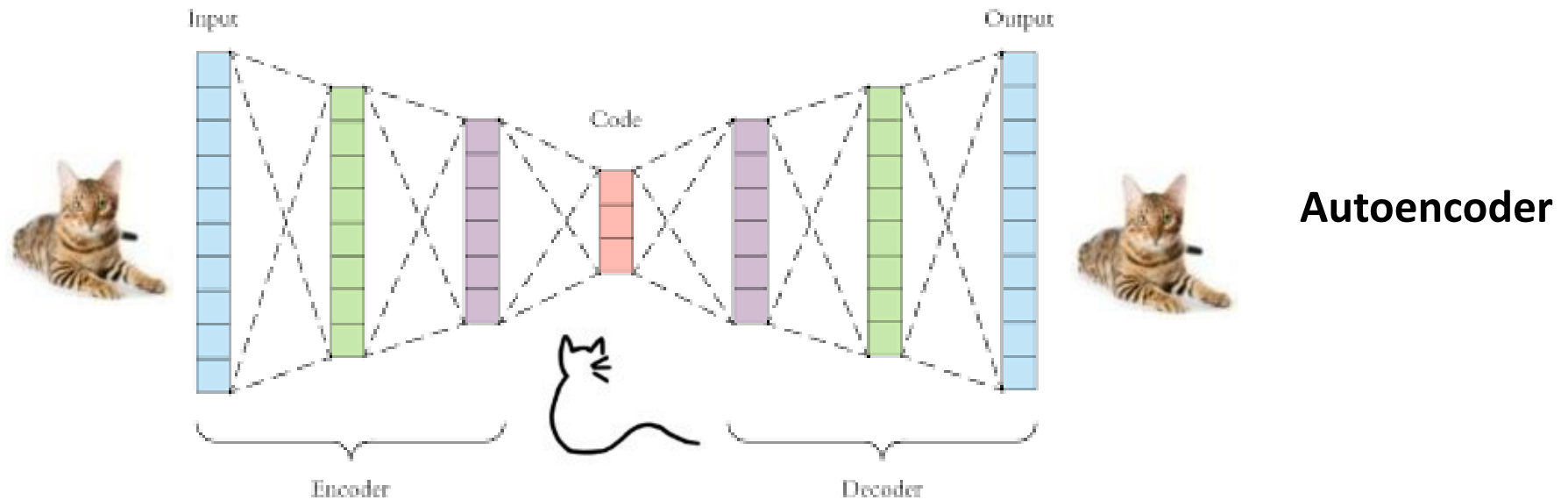


Virtual try-on

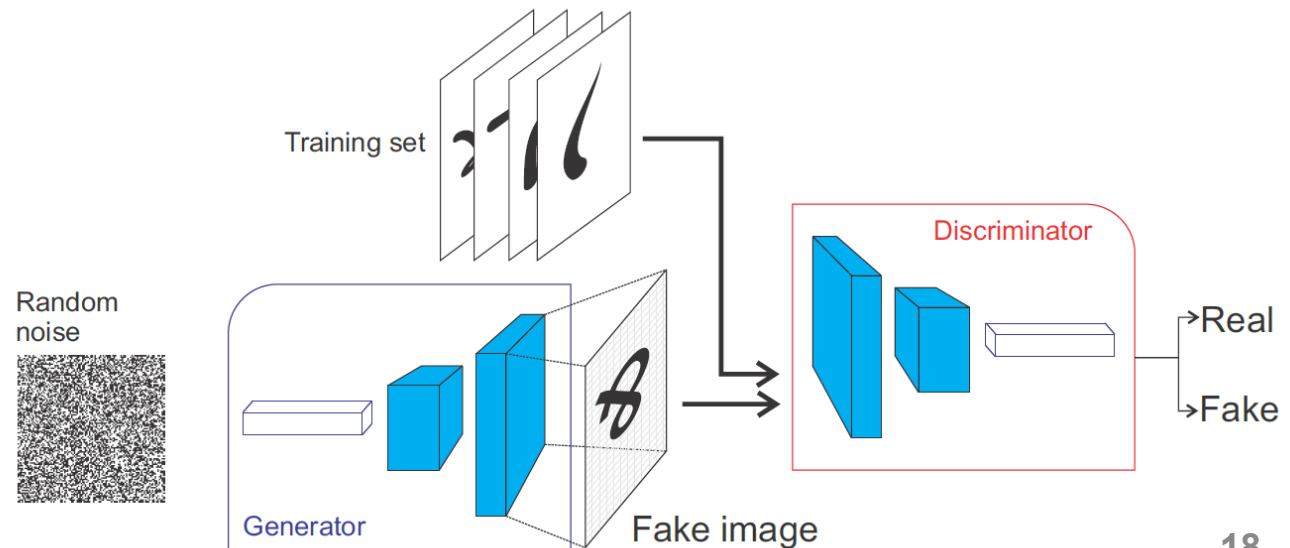


Lee, Sangyun, Gyojung Gu, Sunghyun Park, Seunghwan Choi, and Jaegul Choo. "High-resolution virtual try-on with misalignment and occlusion-handled conditions." In European Conference on Computer Vision, pp. 204-219. Cham: Springer Nature Switzerland, 2022. 17

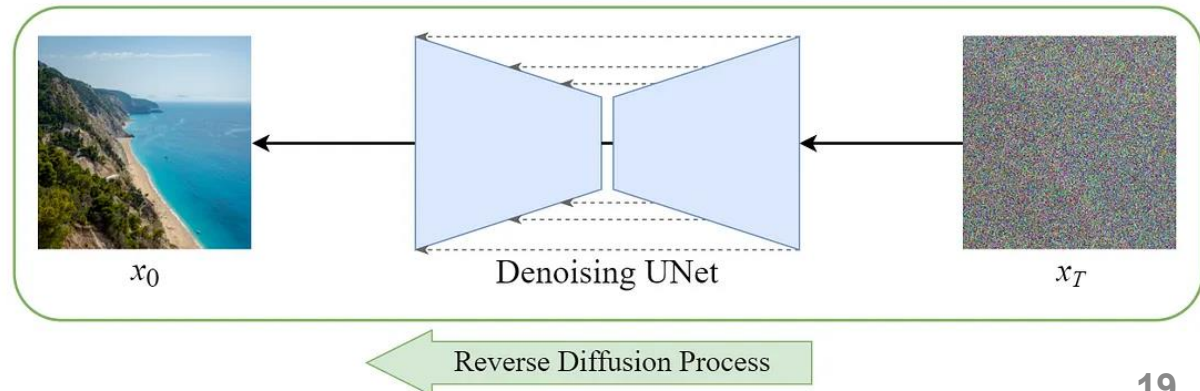
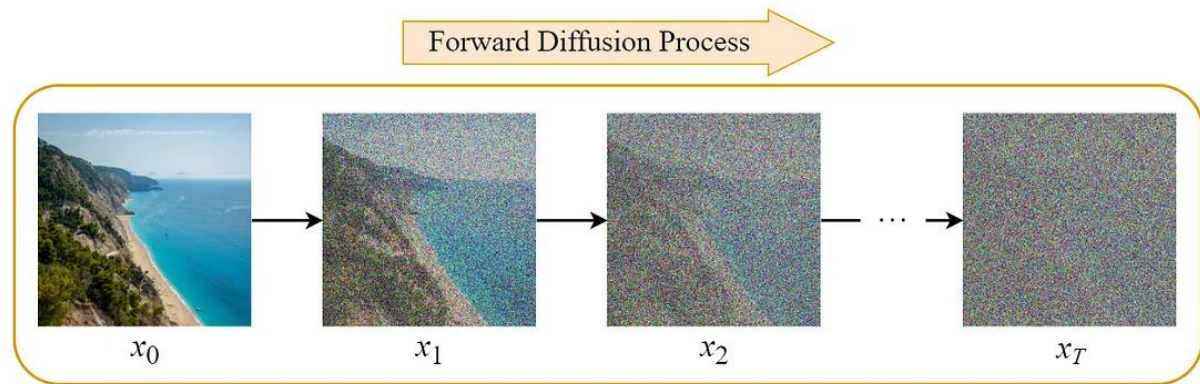
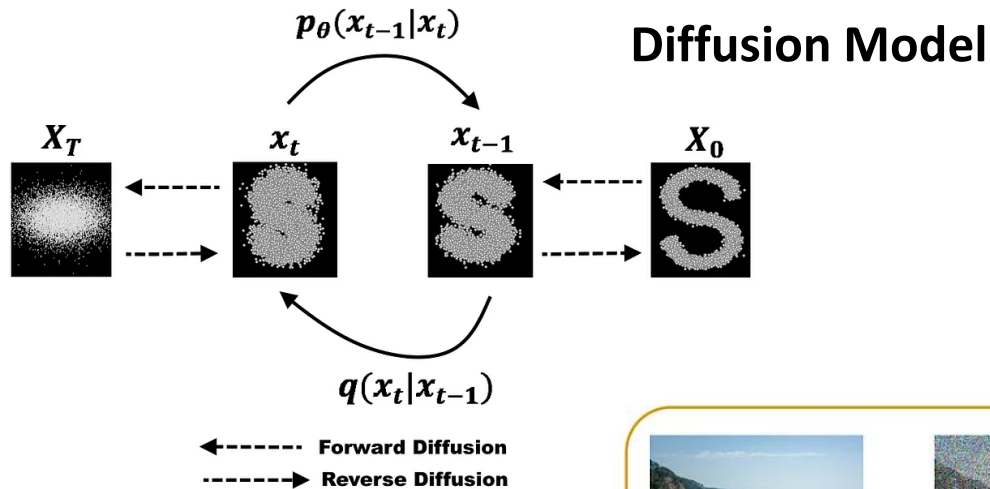
Key techniques in image generation



Generative adversarial network

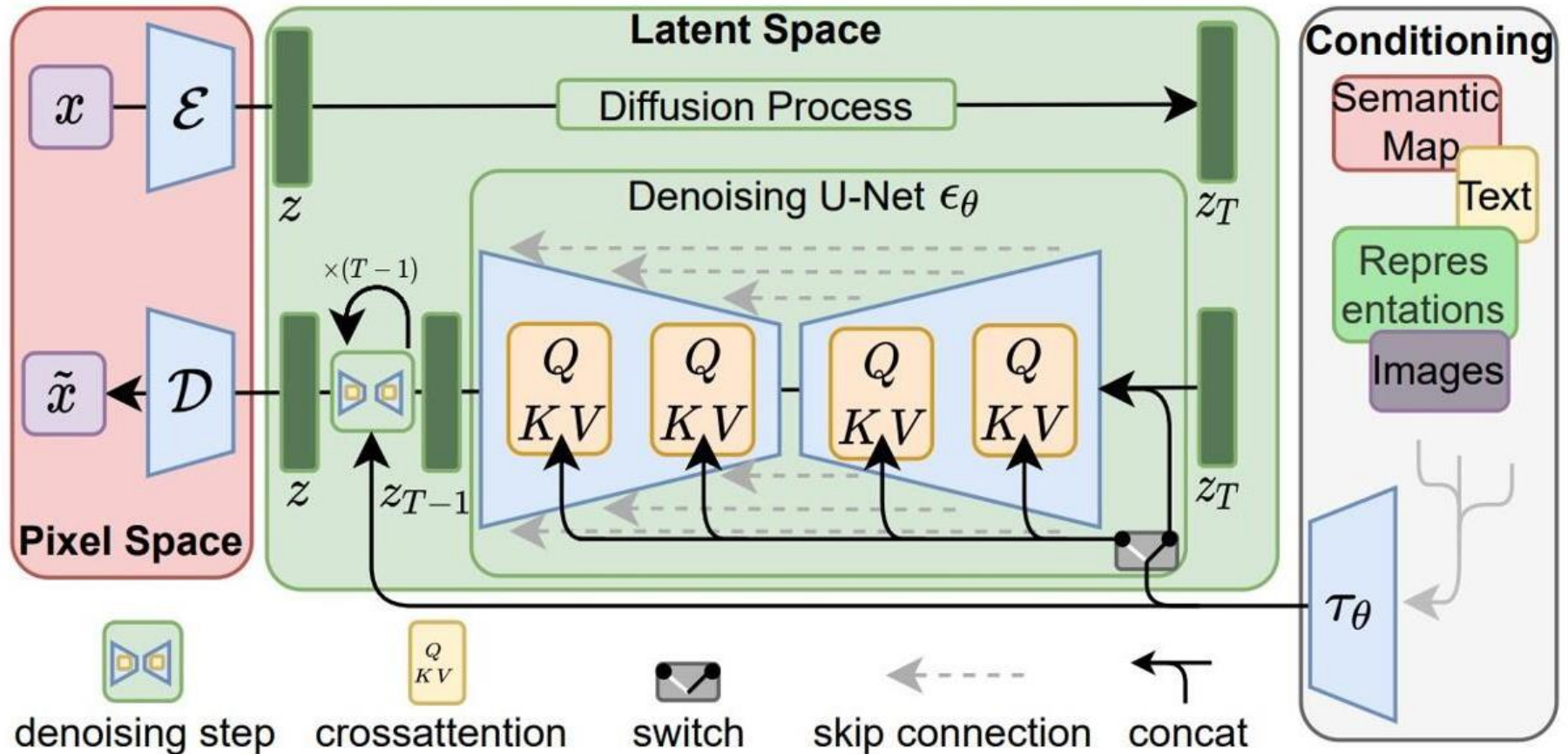


Key techniques in image generation

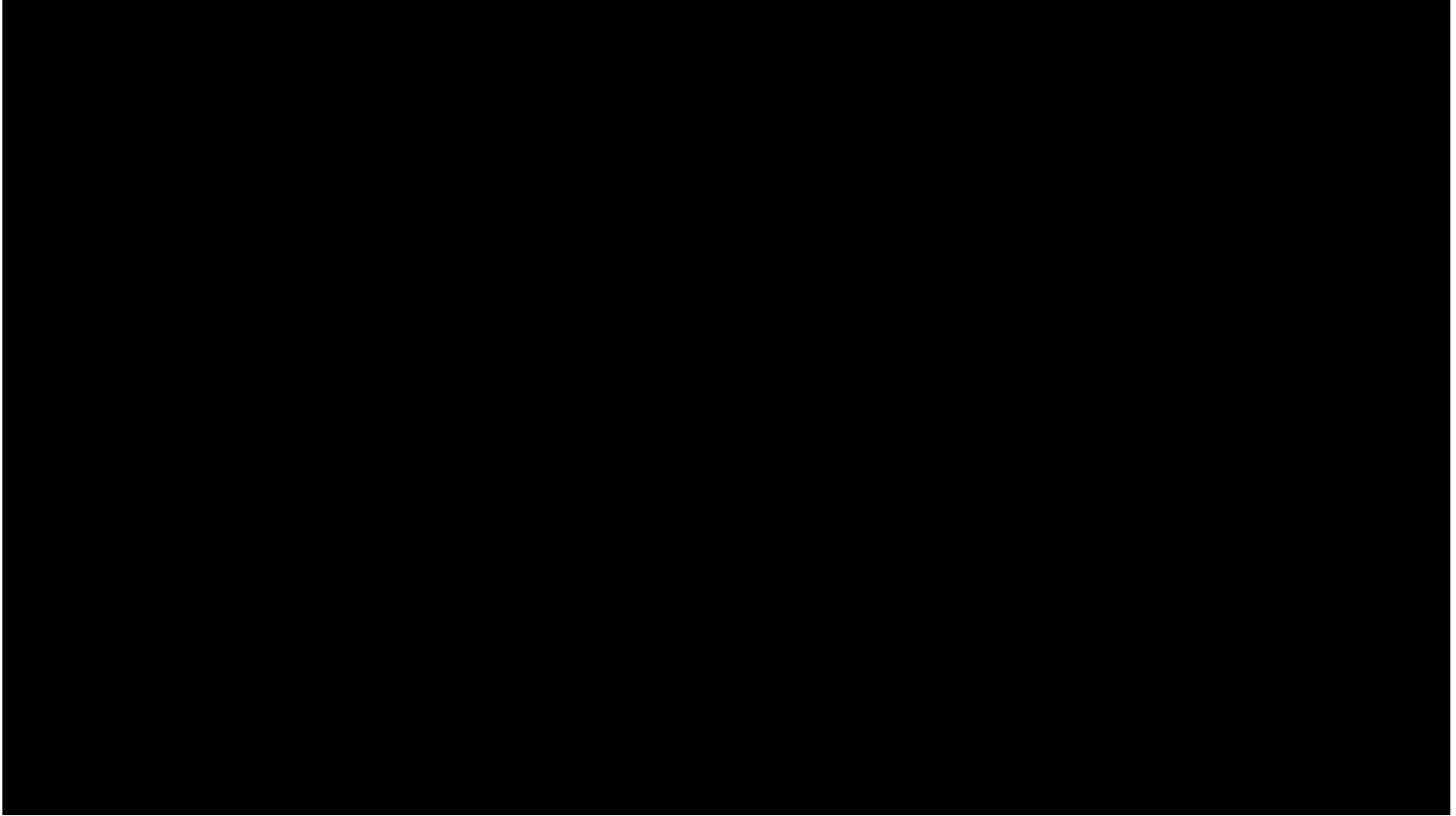


Stable Diffusion Model

Stable diffusion model: An explanation



Video generation: Deepfake



Source: [YouTube video](#)

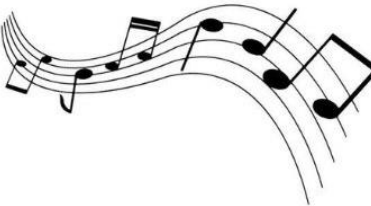
Audio generation: AI music



- Generative AI empowers creators to generate new warbles, chimes, measures, and even entire songs.

 Meta AI
AudioCraft

AWS
DeepComposer

Google
MusicLM 

Virtual Interior Design



Original image

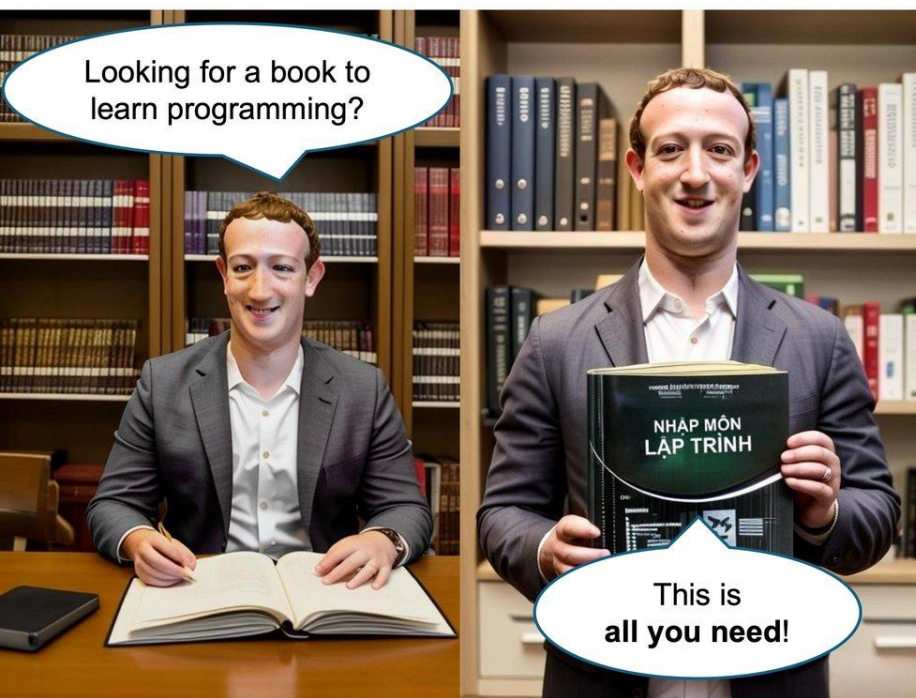


"Japanese style,
wooden wall,
cotton puff"



"A Van Gogh
style painting
on the wall"

Le, Minh-Hien, Chi-Bien Chu, Khanh-Duy Le, Tam V. Nguyen, Minh-Triet Tran, and Trung-Nghia Le. "VIDES: Virtual Interior Design via Natural Language and Visual Guidance." ISMAR 2023.



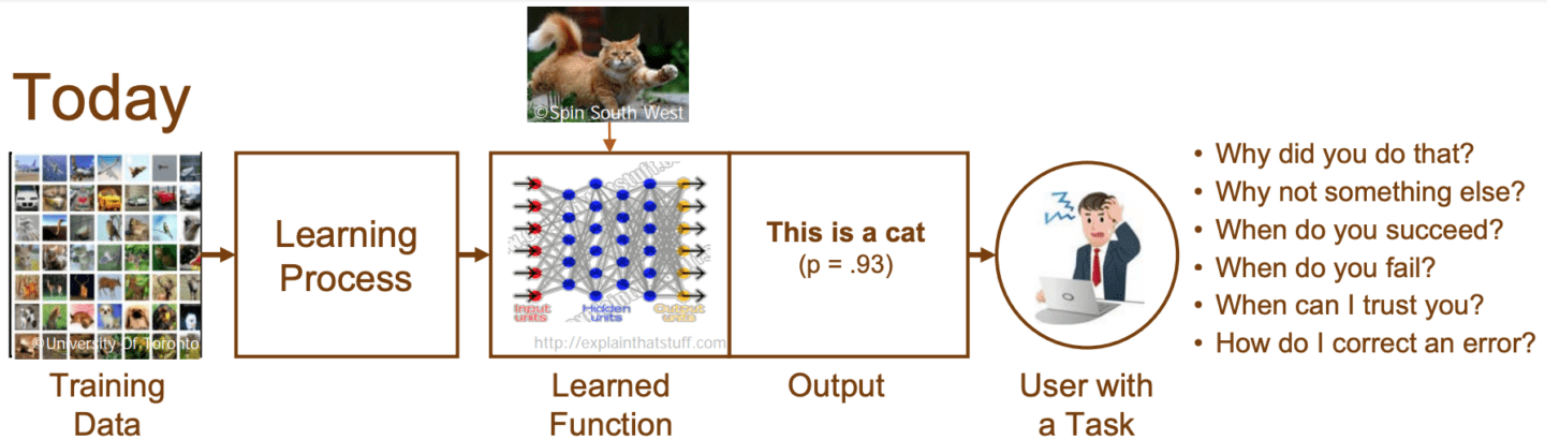
Trong-Vu Hoang, Quang-Binh Nguyen, Minh-Triet Tran, and Trung-Nghia Le. "Generative AI for Human Creativity." Honor program thesis year 2020.

Explainable AI (XAI)

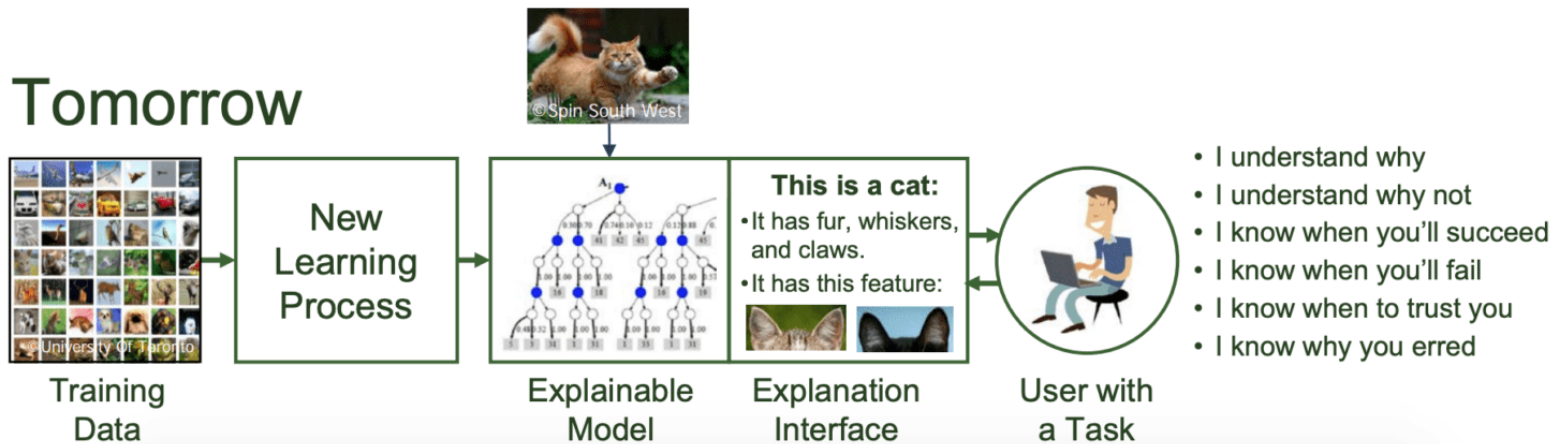
- XAI enables a ML model to provide an explanation for its decisions and action, which can be understood by humans.
 - It contrasts with the "black box" concept in ML where we cannot explain why an AI arrived at a specific decision.
- XAI is desired in use cases involving accountability, fairness and transparency, with the aims of enhancing the trust between humans and machines
- Higher visibility into model decision-making process (which helps with transparency)

Explainable AI (XAI)

Today



Tomorrow



Explainable AI: Challenges

- How to generate explanations that are both **accurate and understandable**.
- XAI models can be more **difficult to be trained and tuned**, compared to non-explainable machine learning models.
- XAI systems are more difficult to deploy since explainability features need some level of **human-in-the-loop intervention**.
 - Explainability is how to take an ML model and explain the behavior in human terms.

Context image



Marked areas in the image were relevant for the answer and hidden areas were irrelevant



What is the man playing?

The AI
answer

soccer (99%)
football (0%)
soccer ball (0%)

XAILab Demo: Explainable VQA

Context image



Marked areas in the image were relevant for the answer and hidden areas were irrelevant



What is the orange object?

The AI
answer

cone (10%)
balloon (8%)
carrot (7%)

XAILab Demo: Explainable VQA

Context image



What flag is it?

Marked areas in the image were relevant for the answer and hidden areas were irrelevant



The AI
answer

american (27%)

usa (15%)

us (6%)

XAILab Demo: Explainable VQA

Acknowledgements

- Some parts of the slide are adapted from
 - Maria-Florina Balcan, Introduction to Machine Learning, 10-401, Spring 2018, Carnegie Mellon University
 - Ryan Urbanowicz, An Introduction to Machine Learning, PA CURE Machine Learning Workshop: December 17, School of Medicine, University of Pennsylvania



...the end.

